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HW- Linear Models Q4

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1. SSE

$$SSE = \sum_i (\hat{y} - y_i)^2$$
$$= \sum (b_0 + b_1 z_{i,1} + b_2 z_{i,2} - y_i)^2$$

2. Partial Derivatives

$$\frac{\partial SSE}{\partial b_0} = \sum 2(b_0^* + b_1 z_{i,1} + b_2 z_{i,2} - y_i) = 0$$
$$\frac{\partial SSE}{\partial b_1} = \sum 2 z_{i,1}(b_0 + b_1^* z_{i,1} + b_2 z_{i,2} - y_i) = 0$$
$$\frac{\partial SSE}{\partial b_2} = \sum 2 z_{i,2}(b_0 + b_1 z_{i,1} + b_2 z_{i,2} - y_i) = 0$$

3. Average error = 0 & $\sum e_i z = 0$

$$\frac{\partial SSE}{\partial b_0} = \sum(\underbrace{b_0 + b_1 z_{i,1} + b_2 z_{i,2}}_{\hat{y}_i} - y_i) = 0 = \sum(\underbrace{\hat{y}_i - y_i}_{e_i}) = 0 = \sum e_i = 0 = \frac{1}{N}\sum e_i = 0 \therefore \bar{e}_i = 0$$

$$\frac{\partial SSE}{\partial b_1} = \sum z_{1,i} \cdot (\hat{y}_i - y_i) = 0 \rightarrow \sum e_i z_{1,i} = 0$$

$$\frac{\partial SSE}{\partial b_2} = \sum z_{2,i}(\hat{y}_i - y_i) = 0 \rightarrow \sum e_i z_{2,i} = 0$$

$$\therefore e_i z = 0$$

4. $b_0^* = \bar{y}$

$$\frac{\partial SSE}{\partial b_0} = \sum b_0^* + b_1 \sum z_{i,1} + b_2 \sum z_{i,2} - \sum y_i = 0 \overset{\text{divide by } N}{\Leftrightarrow} 0 = \frac{n b_0^*}{n} + \cancel{b_1 \bar{z}_1}^{\,0} + \cancel{b_2 \bar{z}}^{\,0} - \bar{y}$$

$$\therefore b_0^* = \bar{y}$$

$$\sum z_{1,i}(\bar{y} + b_1 z_{1,i} + b_2 z_{2,i} - y_i) = 0 \; ; \; \sum z_{2,i}(\bar{y} + b_1 z_{1,i} + b_2 z_{2,i} - y_i) = 0$$

5. Normal Equations

wrt b_0: $n b_0 + b_1(0) + b_2(0) = \sum y_i$

wrt b_1: $\sum b_0 z_{i,1} + \sum b_1 z_{i,1}^2 + \sum b_2 z_{i,1} z_{i,2} = \sum y_i z_{i,1}$

$b_0 \cancel{\sum z_{i,1}}^{\,0} + b_1 \sum z_{i,1}^2 + b_2 \sum z_{i,1} z_{i,2} = \sum y_i z_{i,1}$

$b_0(0) + b_1(\sum z_{i,1}^2) + b_2(\sum z_{i,1} z_{i,2}) = \sum y_i z_{i,1}$

wrt b_2: $b_0 \cancel{\sum z_{i,2}}^{\,0} + b_1(\sum z_{i,1} z_{i,2}) + b_2(\sum z_{i,2}^2) = \sum y_i z_{i,2}$

$A\,b = C$

$$\begin{bmatrix} n & 0 & 0 \\ 0 & \sum z_{i,1}^2 & \sum z_{i,1} z_{i,2} \\ 0 & \sum z_{i,1} z_{i,2} & \sum z_{i,2} \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum y_i z_{i,1} \\ \sum y_i z_{i,2} \end{bmatrix}$$

6. Divide by n & plug in $z_j = x_j - \bar{x}_j$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sum(x_{i,1} - \bar{x}_1)^2}{n} & \frac{\sum(x_{i,1} - \bar{x})(x_{i,2} - \bar{x})}{n} \\ 0 & \frac{\sum(x_{i,1} - \bar{x})(x_{i,2} - \bar{x})}{n} & \frac{\sum(x_{i,2} - \bar{x}_2)^2}{n} \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \frac{\sum y_i}{n} \\ \frac{\sum y_i(x_{i,1} - \bar{x}_1)}{n} \\ \frac{\sum y_i(x_{i,2} - \bar{x}_2)}{n} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & Var(x_1) & Cov(x_1, x_2) \\ 0 & Cov(x_1, x_2) & Var(x_2) \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \bar{y} \\ Cov(y, x_1) \\ Cov(y, x_2) \end{bmatrix}$$

This shows that bivarate linear regression coefficients are tied to the variance of the explanatory variables, covariance between explanatory variables, & covariance between dependent variable & the explanatory variables.